

Abu Jafar Saifullah

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EXPERIENCE

Optimizely – Content Marketing Platform

Dhaka, Bangladesh

Software Engineer II (promoted from SE I, Apr 2026)

Apr 2024 – Present, Full-time

- **Design System Migration:** Drove CMP's multi-year migration from a legacy in-house design library to the **OptiAxiom design system**, authoring 10+ React/TypeScript adapter components and **15+ V1→V2 component migrations** in the design-system repo; authored the official **Axiom Migration Guide** and **Icon Contribution Guide** adopted org-wide across CMP and Opal.
- **Axiom V3 Brand Refresh:** Led the Axiom V3 rollout in CMP including a **module-federation singleton fix** that eliminated duplicate React context errors across micro-frontends; migrated color token system and TreeSelector component.
- **Opal AI App-Companion Integration:** Owned end-to-end integration of Optimizely's Opal AI app-companion SDK – migrated the service from **Express to Hono**, designed the **AWS CodeArtifact publishing pipeline**, built the permissions layer and test suite, and authored the Backpopulation Guide that graduated the feature from flagged to GA.
- **AI Event Instrumentation:** Built the reusable `useAppEvents` React hook and instrumented structured AI-context events across **10+ product surfaces** (Home, Plan, Campaign, Tasks, Workflows, Comments, Fields, Templates, Milestones, Events).
- **Opal Proteus – Block-based AI UI:** Implemented backend Block-context extraction for LLM prompts, frontend Block form interaction, and MCP App resource rendering for Optimizely's AI agent document UI.
- **Spotlight & Status System:** Centralized fragmented per-feature status indicators into a unified **StatusBadge** component backed by a shared server-side schema across campaigns, tasks, and milestones; surfaced in global Spotlight search.
- **Platform Quality:** Resolved duplicate comment/task creation on rapid clicks (server + client), fixed duplicate in-app notifications, refactored Dark Mode to CSS variables, and added 8 missing i18n locale variants.
- **Code Review & Mentorship:** Reviewed **300+ pull requests** across cmp-client, cmp-server, and opal-app; de-facto sign-off owner for design-system migrations and AI-tracking instrumentation.

Orion Informatics Ltd.

Dhaka, Bangladesh

Software Engineer

Jul 2023 – Mar 2024, Full-time

- Built custom graphs and advanced data filters for an Attendance Management System to enhance reporting capabilities.
- Designed and implemented a TradeStation API integration for real-time stock price data; deployed on **Azure** using **.NET**.

SEMERU Lab, William & Mary

Remote

Research Intern, supervised by [Dr. Denys Poshyvanyk](#)

Aug 2023 – Dec 2023

- Built a programming language-specific tokenizer to improve NLP model predictions for software engineering tasks.

EDUCATION

University of Dhaka

Dhaka, Bangladesh

M.Sc. in Software Engineering (Final Semester); **GPA: 3.92/4.00**

Apr 2025 – Aug 2026

B.Sc. in Software Engineering; **GPA: 3.89/4.00**

Jan 2019 – Dec 2023

Notre Dame College

Dhaka, Bangladesh

Higher Secondary Certificate; **GPA: 5.0/5.0**

Jun 2014 – Jun 2016

PUBLICATIONS

Abu Jafar Saifullah, Tasmia Zerin, Zerina Begum, Kazi Sakib. “NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation.” In *Proceedings of the 20th International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE 2025)*. Accepted. [\[GitHub\]](#)

PROJECTS

NavA11y | *Node.js, Playwright, Chromium* | [Code](#) | *Published – ENASE 2026*

- Dynamic WCAG 2.4 accessibility testing tool – the first to cover all five focus-behavior success criteria. Drives a real browser to simulate keyboard navigation and inspect rendered page state at each focus event. Achieved 100% precision and recall on a 22-page labeled dataset; detected 2,947 violations across 26 production websites with 90% confirmed true-positive rate.

RepairA11y | *Node.js, Playwright, Vitest* | *M.Sc. Thesis – In Progress*

- LLM-driven automated repair pipeline for WCAG 2.4 accessibility violations; evaluates whether runtime-evidence prompting outperforms static-evidence (HTML/screenshot) prompting for accessibility patch generation. Uses NavA11y as validation oracle.

Clippy | *Angular, Node.js, AWS S3* | [GitHub](#) | *ICSE SRC 2023 – Top 6*

- Smart PDF reader with knowledge graph extraction, citation network visualization, and paper recommendation. OTP-based auth, Azure CI/CD. 392 commits, 5 GitHub stars; submitted to ICSE 2023 Student Research Competition.

BPE Tokenizer | *Python, Flask, HuggingFace* | [PyPI](#) | *Published Package*

- Byte Pair Encoding tokenizer implemented from scratch and published as `python-bpe-tokenizer` on PyPI; includes a Flask web interface for training on custom corpora and tokenize/detokenize workflows.

Mr. Bin | *MERN, Spring Boot 3, TensorFlow, Maps API* | *JavaFest 2023 (Therap)*

- Full-stack recycling and exchange marketplace with ML-based object detection for item classification; MERN stack with a Spring Boot 3 / Java 17 backend rewrite. Entered in JavaFest 2023 (Therap competition).

EmitXplorer | *React, ASP.NET Core, Python/ML, TensorFlow* | *NASA Space Apps 2023*

- Gamified greenhouse gas visualization platform using SCISAT satellite data and ML-based CO₂ concentration prediction; built for NASA Space Apps Challenge 2023.

API Miners GPT | *ASP.NET Core, React, OpenAI API, EF Core* | *BUET Hackathon – Best UI/UX*

- GPT-powered PDF chatbot with audio I/O, OCR, and Stable Diffusion image generation; .NET backend with SQL Server, React + Tailwind CSS frontend. Won Best UI/UX at BUET Hackathon 2023.

MiniFacebook | *React, Node.js, Docker, NGINX* | [GitHub](#)

- Demonstrated monolith-to-microservices decomposition – split a Node.js app into User, Story, and Status services behind an NGINX reverse proxy / load balancer, orchestrated with Docker Compose.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Frontend: React 18/19, Next.js, Angular, Vite, Tailwind CSS, shadcn/ui, Radix UI, Chakra UI, Framer Motion, Storybook, Module Federation, Rspack

Backend: Node.js, Hono, Express, FastAPI, Spring Boot 3, ASP.NET Core, Entity Framework Core

Testing & Automation: Playwright, Vitest, Karma, Mocha, Chai, Sinon, axe-core, Selenium

Databases: MongoDB, PostgreSQL (Supabase), SQL Server, Firebase, Redis

DevOps / Cloud: AWS CodeArtifact, AWS S3, Azure, Cloudflare Workers, Docker, NGINX, GitHub Actions, Netlify

ML / NLP: TensorFlow, HuggingFace Transformers, BPE Tokenization, TextRank, TF-IDF, scikit-learn

Concepts: Design Systems, Micro-frontends, Dynamic Program Analysis, REST APIs, Microservices, OOP, Design Patterns, Agile/Scrum

AWARDS & ACHIEVEMENTS

Champion – Refactoring Competition (2023) **Best UI/UX** – Hackathon (2023) **Top 6** – ICSE SCORE 2023

Champion – Project Showcasing (2022) **Runners-up** – Project Showcasing (2021) **600+** problems solved on Codeforces & LeetCode